

Urbanization from the Bottom Up

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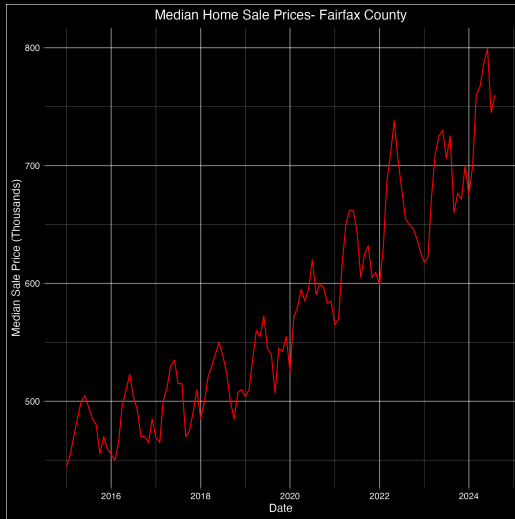
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Outline

1. Empirical motivation: what happened to housing?
2. The NIMBY ABM: model design and mechanics
3. Building with generative AI
4. Parameter sweeps: what we learn overnight
5. Connections and future work

Fairfax County Home Prices



Why Agent-Based Modeling?

Standard models assume:

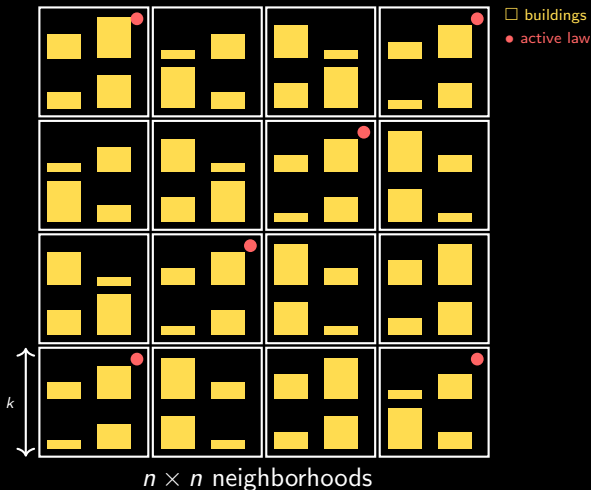
- ▶ Homogeneous agents
- ▶ Perfect information
- ▶ Instantaneous market clearing
- ▶ Zoning as an exogenous constraint

Housing is fundamentally:

- ▶ Heterogeneous (location, type, vintage)
- ▶ Locally governed (neighborhood politics)
- ▶ Path-dependent (who is already there)
- ▶ Endogenous supply (residents vote on zoning)

ABM lets residents, buildings, and laws co-evolve.

City Structure



Default: 10×10 neighborhoods, each 8×8 buildings = **6,400 buildings**.
Building *height* = number of stacked dwellings. Neighbors vote on zoning laws.

Agent Preferences

Each agent i has a job location and residential preferences drawn from log-normal / truncated-normal distributions:

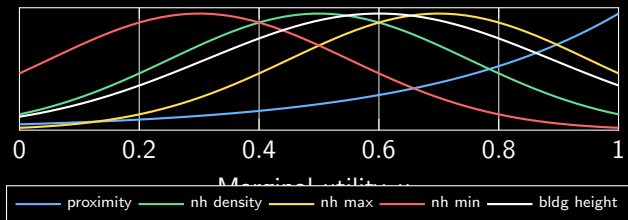
Parameter	Meaning	Distribution
b_i	budget	log-normal
$\bar{d}_i$	preferred neighborhood density	log-normal
$\bar{h}_i^{\max}, \bar{h}_i^{\min}$	preferred max/min heights	log-normal
$\bar{h}_i$	preferred building height	log-normal
$\sigma_i^{nh}, \sigma_i^b$	sensitivity to deviations	log-normal
λ_i	proximity decay scale	log-normal
θ_i	Frank copula dependence	log-normal

Job buildings assigned at creation, weighted by $w(b) = \text{height}(b) + 1$.

The Utility Function

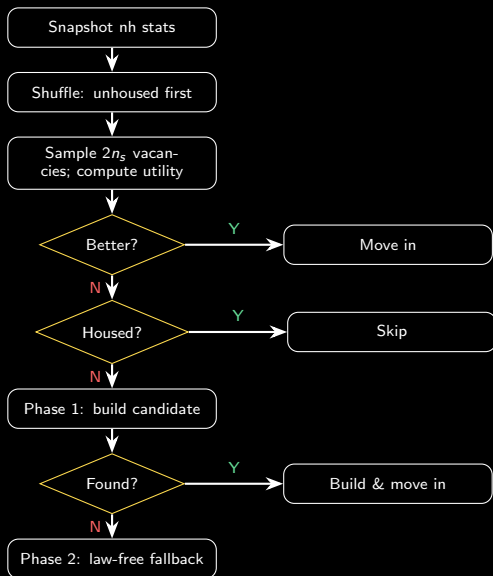
Five marginal utilities $u_k \in [0, 1]$, combined via Frank copula:

$$C_{\theta}(u_1, \dots, u_5) = -\frac{1}{\theta} \ln \left(1 + \frac{\prod_k (e^{-\theta u_k} - 1)}{(e^{-\theta} - 1)^4} \right)$$



$\theta > 0$: positive dependence — all dimensions must be satisfied jointly.

Simulation Step

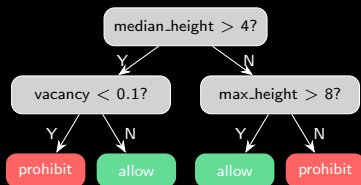


The NIMBY Mechanism

Every T steps:

1. Propose a random depth-2 decision tree over neighborhood observables
2. Deep-copy city: **with law** vs **without law**
3. Run H steps of new inflow on each branch (same RNG seed)
4. Incumbent residents compare utility across branches
5. Majority vote: if $> 50\%$ prefer the law branch $\Rightarrow$ law adopted

Laws are **conditional**: the same law may allow building in one period, block it in another.



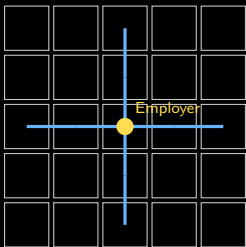
Transport Network & Employer

Roads (BFS hop cache):

- ▶ Neighborhoods connected by a road network
- ▶ Effective distance
= $\min(\text{Euclidean}, d_{\text{subway}})$
- ▶ Subway = walk to hub + hops
 $\times$ road unit + walk from hub
- ▶ Greedy road planner adds
highest-contrast edges

Employer (branch simulation):

- ▶ Single employer tracks all
worker IDs
- ▶ Every T_e steps: sample k
candidate buildings
- ▶ Deep-copy and run H_e steps on
each candidate
- ▶ Relocate to minimise mean
effective commute



What Emerges

1. **Sorting** — agents self-segregate by density preference
2. **Height cascades** — successful tall buildings attract similar agents, law proposals struggle
3. **Law persistence** — once a restrictive law passes, neighborhood stays low-density
4. **Employer drift** — job center migrates toward high-density worker clusters
5. **Unhoused fraction** — regulated cities produce persistent homelessness even with empty land

Building with Generative AI

≈1,500 lines of Julia + Python + HTML across 10 files

Real-time WebSocket visualization, branch-simulated voting, BFS
road cache

Built over ~9 sessions with Claude as collaborator

How AI Was Used

Architecture

- ▶ Struct design (City, Agent, Law, Employer)
- ▶ WebSocket protocol between Julia $\leftrightarrow$ Blender $\leftrightarrow$ browser
- ▶ Frank copula parameterisation

Implementation

- ▶ Boilerplate (BFS, JSON serialisation)
- ▶ Debugging async race conditions
- ▶ Blender mesh geometry API

What AI does not replace

- ▶ Model design choices (what *should* agents optimize?)
- ▶ Identifying economically meaningful bugs
- ▶ Interpreting simulation output
- ▶ Deciding which parameters to sweep

A Concrete Example: The Employer Bug

1. Employer initialized at building ID 1 — grid corner (0,0)
2. All 5,000 agents assigned job there
3. Agents clustered at corner; single tower > 200 floors
4. AI spotted the bug when shown the output: “employer should start at the city centroid”
5. Fix: compute nearest building to $\left(\frac{n_x \cdot k}{2}, \frac{n_y \cdot k}{2}\right)$

The bug was *economically obvious* in the output but invisible in the code. AI found it by reasoning about *intent*, not just syntax.

What Worked and What Required Care

Worked well

- ▶ Iterative refinement: “make roads thin and black”
- ▶ Explaining intent, not code
- ▶ AI as a persistent memory of the codebase
- ▶ Catching off-by-one and async bugs

Required human judgment

- ▶ Verifying economic logic (copula signs, utility direction)
- ▶ AI over-engineers without constraints
- ▶ Hallucinated API calls (Blender Python)
- ▶ Deciding what the model is *for*

AI accelerates implementation; it does not replace the researcher.

Research Questions for the Sweep

1. How does **preference correlation** (θ) affect law adoption rates?
2. Does a **longer voting horizon** (H) produce more or less restrictive cities?
3. When does the city tip from **low-density equilibrium** to **high-density equilibrium**?
4. What is the unhoused fraction as a function of law frequency?

Parameter Sweep Design

Parameter	Meaning	Values	Axis
θ_μ	copula dependence	0.5, 2, 5, 10	A
H	law eval horizon	5, 10, 20	B
T	law eval every N steps	5, 10, 20	C
σ_μ^{nh} (log)	density sensitivity	-1, -0.5, 0, 0.5	D

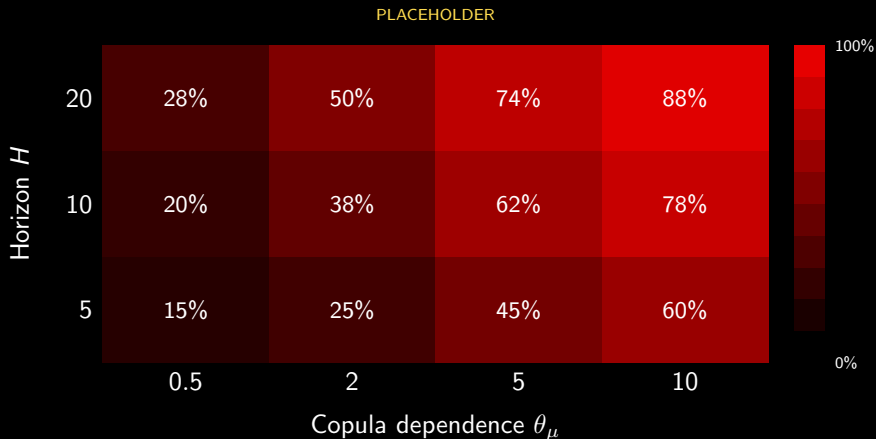
Focused design: sweep axes $A \times B$ ($4 \times 3 = 12$ cells)

$\times 3$ replications $\times 200$ steps each = **36 runs overnight**

Fix $T = 10$, $\sigma_\mu^{nh} = -0.5$ (tight preferences); vary θ and H .

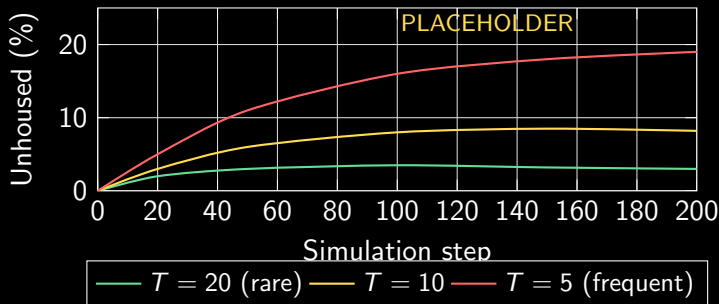
Outcome variables: % neighborhoods with active law, mean height, unhoused %, employer displacement.

Expected Result: Law Adoption Heatmap

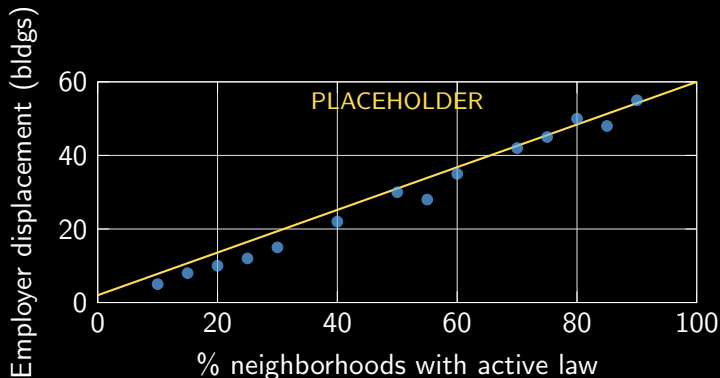


Higher θ and longer horizon $\Rightarrow$ more restrictive laws.

Expected Result: Unhoused Fraction vs. Law Frequency



Expected Result: Employer Displacement vs. Law Density



More laws $\Rightarrow$ employer pushed further from initial location.

Connection to the Real Market

1. **NIMBY** mechanisms protect incumbent interests at the neighborhood scale:
incumbents vote to restrict new supply, preserving neighborhood character and property values
2. This produces **supply restriction** $\Rightarrow$ elevated prices + reduced volume
3. ABM captures the **feedback**: laws slow supply $\Rightarrow$ prices rise $\Rightarrow$ incumbents have more to protect $\Rightarrow$ laws tighten
4. Fairfax County is a natural calibration target: documented high regulatory burden, post-2021 price stickiness

Zoning Laws Are Already Trees

An experiment: fed the full Fairfax County zoning ordinance to ChatGPT and asked:

“To what extent can these rules be represented as decision trees?”

- ▶ **Most can** — permit thresholds, setbacks, height limits, and use-by-right tables all branch on observable lot and neighborhood attributes
- ▶ **Main exception:** special-purpose and planned-development districts, which require case-by-case negotiation

Why this matters computationally:

- ▶ Zone types map naturally to Julia **structs**
- ▶ **Multiple dispatch** lets different law types share a single `can_build_in_neighborhood` interface — no if/else towers
- ▶ Opens a path to a **synthetic Fairfax County**: parameterise the model directly from the ordinance, match the observed height and vacancy distribution

Future Work

Model extensions

- ▶ Heterogeneous employers (job clusters, not a single center)
- ▶ Realtor agent coordinating buyer/seller matching
- ▶ Demographic aging (cohort-based entry/exit)

Calibration

- ▶ Fairfax County parcel data $\Rightarrow$ building height distribution
- ▶ HMDA data $\Rightarrow$ agent budget distribution
- ▶ Zoning board records $\Rightarrow$ law adoption frequency
- ▶ Moment-matching on median price trajectory

Summary

1. **Empirically**: NIMBY zoning explains post-2022 price stickiness in tight markets
2. **The ABM**: heterogeneous agents sort, build, vote on laws, and drive employer location — rich emergent dynamics
3. **AI as collaborator**: accelerates implementation; human judgment still required for model design and interpretation
4. **Parameter sweeps**: higher preference correlation and longer voting horizons both amplify law adoption and unhoused fractions
5. **Next step**: deepen the bridge between model mechanics and real urban policy

Thank You

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Code: github.com/jsschuler/urban1

Questions welcome on: ABM design · AI-assisted research · parameter
sweep results